

Arnab Mandal

+91-9560452773 | arnabmandal2912@gmail.com | [linkedin.com/arnabmandal2912](https://www.linkedin.com/arnabmandal2912) | github.com/Eros483 | [Portfolio](#)

EDUCATION

Shiv Nadar University

B.Tech in Computer Science Engineering (8.59/10.0)

Noida, Delhi-NCR, India

Aug 2023 - Jul 2027

Heera Lal Public School, Class XII

Percentage : 91%

Madan Pur Dabas, Delhi-NCR, India

May 2023

Navy Children School, Class X

Percentage : 97%

Visakhapatnam, Andhra Pradesh, India

May 2021

INTERNSHIPS & EXPERIENCE

Intern, Enterprise AI Platform, Fidelity International, Gurgaon

Jun 2026 - Current

- Cut **PII** redaction latency by **92%** via **AWQ**-quantized fine-tuning of Gemma 4 on **SageMaker**, reaching **96% F1**.
- Shipped a document-sensitivity guardrail for the internal AI gateway, monitoring **15k+** uploads/month via **ELK**.
- **Business Impact:** Cut manual audit effort by **40%** and redaction costs by **85%**.

AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati

Dec 2025 - Jan 2026

- Optimized Llava 1.5 Pipeline serving **50+** clients by migrating inference to TensorRT alongside Triton inference server.
- **Business Impact:** Achieved GPU utilization increase by **30%** and serving **8x** clients with no additional resources.

AI Intern, Critical AI Pvt Ltd, Bengaluru

Jul 2025 - Sep 2025

- Deployed multimodal RAG pipelines (FAISS, llama-cpp, OSM, TTS, STT, MongoDB) behind FastAPI microservices.
- **Business Impact:** Added **3+** features and transitioned to on-prem LLM stack, reducing inference cost by **30%**.

AI Intern, WESEE, Ministry of Defence, New Delhi

May 2025 - Jul 2025

- Integrated MCP agents for SQL-based analytics and fine-tuned YOLOv8 pipelines (**94%** mAP@50, **68%** mAP50-95).
- **Business Impact:** Cut SME report prep time by **~90%** leveraging **prompt engineering**.

PROJECTS

Open-Source: Domino Speculative Decoding for vLLM (85k★) 🔄 🧠

- Implemented Inference and Distributed training support for Domino, reporting a **18%** throughput gain on Qwen3.
- Improved acceptance length by **27%** over DFlash across 7 benchmarks, including **GSM8K** and HumanEval.
- Technology Stack: Python, PyTorch, Transformers

Graph-RAG Text-to-SQL Agent for Enterprise Databases 🔄 📄

- Engineered agent for a healthcare analytics platform, tested over **43** tables and **62** relationships in a **1GB** warehouse.
- Boosted multi-join query execution accuracy by **33%** and validated by **CEO+CTO** for **production** adoption.
- Technology Stack: Python, LangGraph, FAISS, NetworkX, FastAPI

AutoLinks: Distributed Semantic Internal Linking Engine for SEO 🔄 📄

- Architected async, autoscaling Goroutine workers with Redis-backed job queue + DLQ for bursty workloads.
- Utilized LLM-as-a-judge, achieving **85+%** acceptance, with **p95** Latency of **600ms**, lowering Gini coefficient by **20%**.
- Technology Stack: Go, chi, GLiNER, trafilatura, Qdrant

TECHNICAL SKILLS

Languages	Python, SQL, Go, TypeScript, C++, Cuda
AI & ML	PyTorch, Transformers, LangChain, LangGraph, Optuna, vLLM, LiteLLM
DevOps & MLOps	FastAPI, chi, Kubernetes, Jenkins, Docker, MLflow, Triton, TensorRT, LangSmith
Cloud	GCP, AWS, Azure, Terraform

ACHIEVEMENTS

- **Smart SNU Hackathon'25 Winner:** Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- **LeetCode Knight:** Achieved a **1,925** peak contest rating and solved **400+** DSA problems.
- **Dean's list: Monsoon'23**
- **Semifinalist speaker at SMVIT Parliamentary Debate'24**